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Research on the mechanical properties and damage mechanisms of 3D printed rice husk ash–crumb rubber concrete

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ABSTRACT

The integration of waste crumb rubber in 3D concrete printing offers significant advantages in terms of sustainability, lightweight, and ductility. However, it often results in a substantial decline in mechanical performance and severe anisotropy. This study proposes an eco-friendly 3D-printed rice husk ash-crumb rubber concrete (3DPRRC) by utilizing ground rice husk ash (RHA) as a partial cement replacement (10%, 20%, and 30% by mass). The printability, mechanical properties, damage evolution, and carbon emission assessment of 3DPRRC were systematically evaluated. The experimental results demonstrated that 3DPRRC is printable and exhibits excellent buildability. Notably, the incorporation of RHA significantly enhances the compressive and flexural strengths, with the 20% replacement level yielding the optimal mechanical performance. A critical finding is that RHA preferentially strengthens the interlayer bonding in the traditionally weakest loading direction, thereby effectively mitigating the mechanical anisotropy inherent in 3D-printed structures. SEM test results indicate that RHA enhances the mechanical properties of 3D-printed crumb rubber concrete (3DPRC) by promoting the formation of hydration products. Digital Image Correlation (DIC) and Acoustic Emission (AE) analyses further elucidated the reinforcement mechanisms, revealing that RHA enhances interlayer bonding strength. This research provides a promising pathway toward sustainable and high-performance additive manufacturing in construction.

1. Introduction

The global construction industry is currently facing a critical dual challenge: the urgent need to decarbonize traditional cementitious materials and the transition toward automated, high-efficiency manufacturing via 3D concrete printing (3DCP). Concurrently, the exponential expansion of the global transportation sector has resulted in a massive accumulation of end-of-life tires (ELTs), posing serious environmental and ecological concerns. While China’s 14th Five-Year Circular Economy Development Plan targets a 78% recycling rate for waste rubber by 2025, a significant disparity remains compared to the near-total recycling achieved in several European nations (e.g., a 100% recycling rate in the Swiss Confederation). Currently, a large quantity of ELTs is disposed of via incineration or landfilling, leading to profound ecological pollution. The recycling of waste tires into rubber particles or powder for incorporation into concrete serves as an effective strategy to address these ecological challenges [1–3], directly contributing to China's strategic goals of reaching a carbon peak and achieving carbon neutrality. Rubberized concrete (RC) has emerged as an eco-friendly [4,5], lightweight construction material characterized by superior durability [6–8], thermal insulation [9], and fatigue resistance [10,11]. Despite these advantages, researchers have found that weak interfacial bonding between rubber particles and cementitious matrix compromises the mechanical properties of RC [12–14].

Traditional concrete casting methods continue to face limitations, including outdated production techniques, high labor requirements, and low operational efficiency [15–18]. Compared to conventional construction methods, 3D concrete printing technology (3DCPT) offers significant advantages in terms of automation potential, design flexibility, construction efficiency, and cost predictability [18–21]. Consequently, 3D printed concrete has been prioritized in national policies and identified as a key research focus in multiple countries. For instance, in 2022, the Chinese Ministry of Industry and Information Technology issued its first list of representative additive manufacturing applications,

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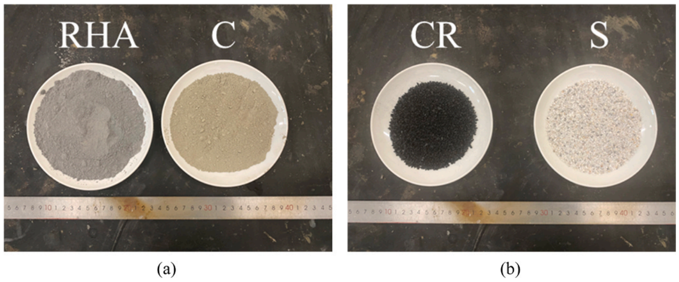


Fig. 1. Cementitious material and aggregate: (a) Cementitious materials and (b) Fine aggregate.

highlighting the integrated production of building structures as a priority area [22]. In the same year, the U.S. Department of Energy awarded $39 million to 18 research teams, with two projects focusing on large-scale 3DCPT development aimed at minimizing resource waste [23]. Several European countries have also accelerated research and implementation of 3DCPT [24].

Although 3D-printed buildings have demonstrated scalability [25, 26], most remain limited to structurally simple landscape features [27, 28] or low-rise housing [29–31]. The large-scale deployment of 3D-printed concrete (3DPC) faces two major challenges: first, excessive cement content resulting from the absence of coarse aggregates compromises eco-efficiency [32]. Second, the layer-by-layer deposition process introduces weak interlayer and intralayer interfaces, resulting in reduced structural strength and pronounced mechanical anisotropy [33–36]. These challenges underscore the need for sustainable, high-performance materials specifically tailored for 3DCPT.

3D-printed rubberized concrete (3DPRC) facilitates sustainable development by combining digitalization with waste valorization [4,5]. Compared to conventional 3DPC, 3DPRC exhibits lower density [37], reduced thermal conductivity, and enhanced acoustic damping properties [38]. However, the mechanical performance of 3DPRC is penalized by a "double-interface" weakening effect: the microscopic interfacial transition zone (ITZ) surrounding the crumb rubber (CR) and the macroscopic interlayer interfaces created during printing. While fiber incorporation [5,39,40] and rubber surface modifications [41–43]. It has been employed to mitigate these weaknesses, the high cement intensity of these mixtures remains a critical sustainability challenge. Therefore, there is an urgent need to explore green, supplementary cementitious materials (SCMs) that can replace cement and enhance the performance of 3DPRC.

The incorporation of pozzolanic mineral admixtures is a widely adopted strategy to optimize 3DPC [9,32]. Among these, RHA has emerged as a renewable supplementary cementitious material (SCM) owing to its exceptional pozzolanic activity and long-standing interest in cementitious systems [44,45]. As the world's largest rice producer, China generates RHA as a primary byproduct of rice processing; this biomass is abundant, easily collected, and remains underutilized. Without proper valorization, massive RHA waste causes severe environmental pollution [46,47]. Compared to conventional SCMs such as fly ash and silica fume, RHA offers a green and renewable alternative with a favorable performance-to-cost ratio [45,48]. RHA and CR share critical parallels as industrial/agricultural solid waste that requires unconventional disposal. Integrating RHA into 3DPRC as a partial cement replacement addresses the dual challenges of RHA waste management and cement overuse while simultaneously densifying the matrix to mitigate mechanical weaknesses.

3DPRC exhibits substantial potential in waste rubber recycling and advancing digitalization of the construction industry. However, the strengthening mechanisms and degradation pathways of RHA in 3DPRC remain underexplored. In this study, CR and RHA are incorporated to replace quartz sand and cement respectively, with 10 vol% of quartz sand substituted by rubber particles serving as the baseline 3DPRC formulation. The effects of variable RHA replacement levels (0%, 10%, 20%, 30%) on printability, compressive strength, and flexural strength of 3DPRC are systematically investigated, while digital image correlation (DIC) and acoustic emission (AE) monitoring are employed to characterize crack propagation patterns and failure mechanisms.

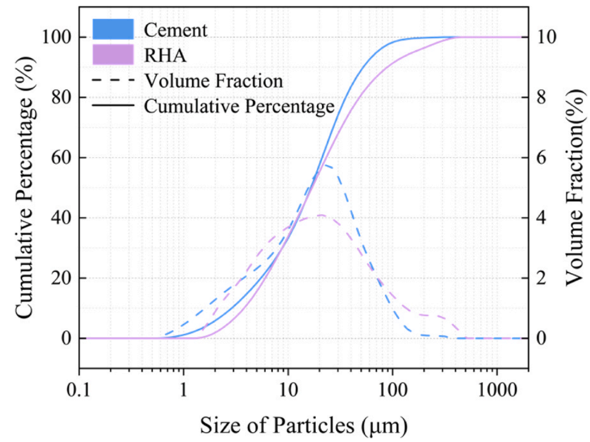


Fig. 2. Particle size distributions of cement and RHA.

2. Materials and experimental methods

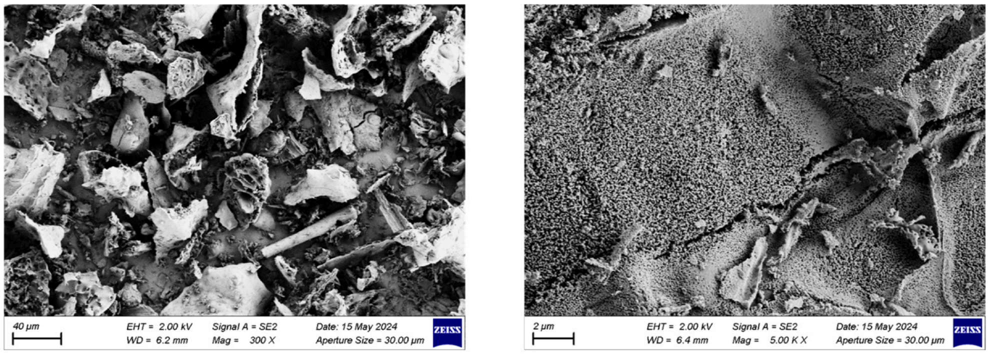
2.1. Raw materials and mixed proportions

The cementitious binder materials (Fig. 1a) consisted of P. O. 42.5 Portland cement (C) and RHA. The RHA used in this study was sourced from a supplier in Jiangsu Province, China. As illustrated in Fig. 1(a), the as-received RHA exhibited a dark gray appearance. The fine aggregate (Fig. 1b) consisted of quartz sand (S) and CR particles. To achieve a well-graded aggregate system, three fractions of quartz sand (10–20, 20–40, and 40–80 mesh) were blended at a mass ratio of 2:1:1, resulting in a medium gradation with a fineness modulus of 2.9. The crumb rubber particles had a particle size range of 1–2 mm. A polycarboxylate-based superplasticizer (PS) was used as the sole admixture to ensure the printability of the mixtures. The particle size distributions of the cementitious precursors are illustrated in Fig. 2.

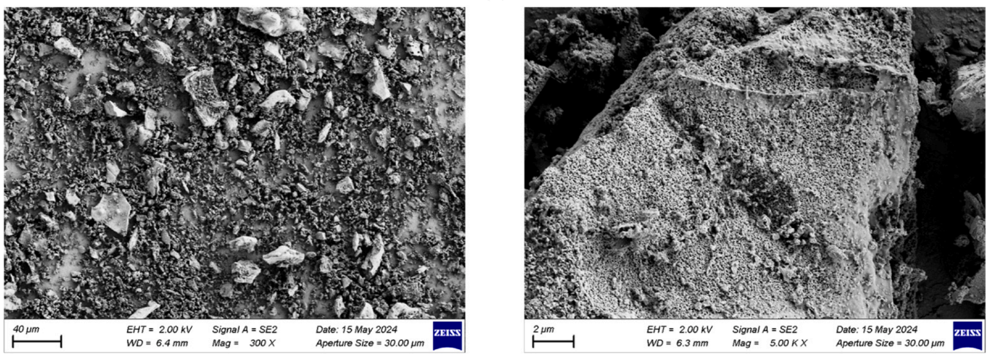
Microstructural characterization using scanning electron microscopy (SEM) (Fig. 3a) revealed that the raw RHA particles exhibited irregular and diverse morphologies with relatively large and uneven particle sizes. The material contained numerous micrometer-scale macropores and nanometer-scale micropores, resulting in a high-water absorption capacity. To enhance its performance, RHA was subjected to mechanical grinding using a ball mill for 50 min. The SEM micrographs of RHA following the grinding process (Fig. 3b) revealed a significant reduction in particle size, with micron-scale macropores largely collapsed. Consequently, this process led to a significant increase in the specific surface area from 451.0 to 552.4 m^2/kg. The increase in surface area and disruption of porous structures improved the pozzolanic reactivity of RHA. Moreover, the refined particle size distribution contributed to a filler effect, which enhanced the particle packing density and, to some extent, improved the workability of fresh concrete mixture.

The chemical compositions of the cement and RHA were determined via X-ray fluorescence (XRF) spectroscopy, with the results summarized in Table 1. RHA was predominantly composed of silicon dioxide (SiO_2), which accounted for 88.49% of its total mass. As illustrated by the X-ray diffraction (XRD) patterns in Fig. 4, the SiO_2 in the RHA existed primarily in an amorphous state, which is indicative of high pozzolanic reactivity.

The detailed mix proportions for the concrete are presented in Table 2. A constant water-to-binder (w/b) ratio of 0.3 was maintained across all experimental groups. To investigate the influence of



(a)



(b)

Fig. 3. Microscopic morphology of RHA: (a) Microstructure of raw RHA and (b) Microstructure of ground RHA.

Table 1
Chemical composition (wt%) of cement and RHA.

Composition	SiO_2	CaO	Al_2O_3	Fe_2O_3	MgO	K_2O	P_2O_5	TiO_2	Loss
Cement	20.08	59.50	6.90	2.77	3.79	1.49	0.13	0.48	3.62
RHA	88.49	1.64	0.21	0.71	0.64	5.32	1.24	0.02	7.22

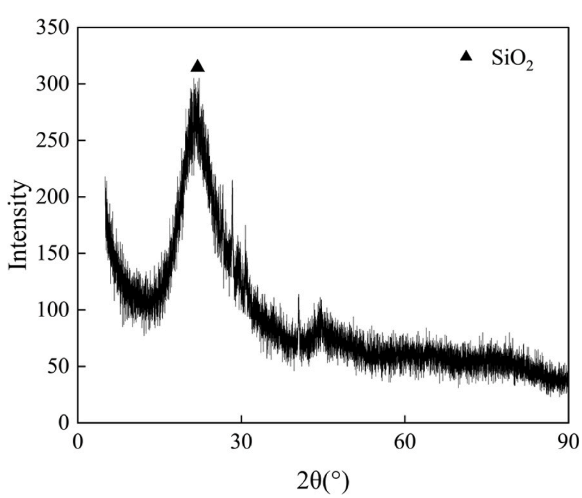


Fig. 4. XRD pattern of RHA.

Table 2
Mixture proportions of concrete (kg/m^3).

Group	C	RHA	S	CR	Water	SP
R0	900	0	810	39.6	270	1.8
R10	810	90	810	39.6	270	3.7
R20	720	180	810	39.6	270	5.1
R30	630	270	810	39.6	270	7.6

Note: Ra: a% mass replacement of cement with RHA.

Table3
Mixing procedure.

Step	Description
1	Mixing Quartz sand and rubber particles at normal for 60 s.
2	Cement and half of the RHA were added and mixed for 120 s.
3	Adding water and SP and mixed at high speed for 180 s.
4	The remaining half of the RHA was uniformly added and mixed for 180 s to ensure complete homogenization.

sustainable additives, CR was utilized to replace 10% of the quartz sand by volume. Furthermore, RHA was incorporated as a partial cement replacement at mass fractions of 0%, 10%, 20%, and 30%. To ensure that the mixture met the stringent rheological requirements for 3D printing, a high-performance polycarboxylate-based superplasticizer (SP) with a water-reduction capacity exceeding 27% was employed. The dosage of SP was the sole parameter adjusted to calibrate the extrudability and buildability of the various mixtures.

Due to the significant water absorption capacity of RHA and the low w/b used in this study, half of the RHA was first removed before dry mixing to facilitate the mixing process. The remaining dry materials were mixed first, followed by the addition of all the water and superplasticizer to the mixture, which was then stirred for 180 s. Finally, the

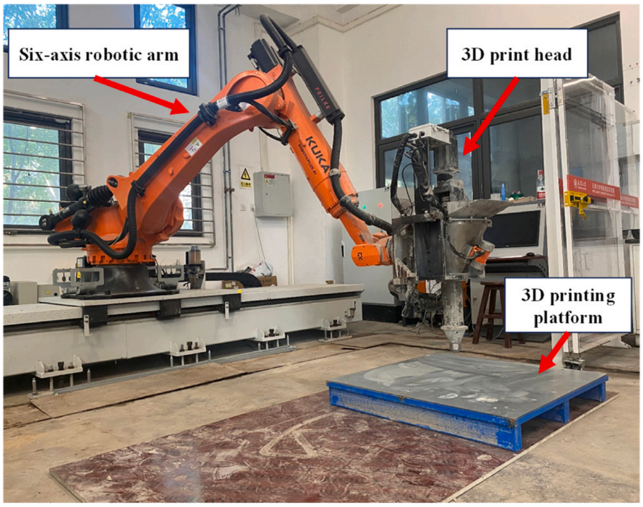


Fig. 5. Six-axis robotic arm and 3D print head.

Table 4
3D printing parameters.

Printing mode	Nozzle diameter	Printing speed	Extrusion width	Layer height
Parallel printing	D= 20 mm	50 mm/s	30 mm	15 mm

previously removed RHA was uniformly reintroduced into the already mixed paste, and the mixture was further stirred for an additional 180 s to ensure complete homogenization. The mixing procedure is outlined in Table 3.

The fresh concrete mixture was divided into two groups for 3D printing and conventional casting. The printed specimens were immediately covered with a plastic film and cured with water for 3 d. Once sufficient strength was attained, the specimens were cut to the required dimensions and transferred to a standard curing room until the 28-day testing age. The cast specimens were prepared by placing the fresh mixture into molds and utilizing mechanical vibration to ensure full compaction. These specimens were covered with film and cured in-situ for 24 h prior to demolding. Following demolding, they were transferred to the standard curing room for an identical 28-day period, after which mechanical testing was conducted.

2.2. 3D printing system and process

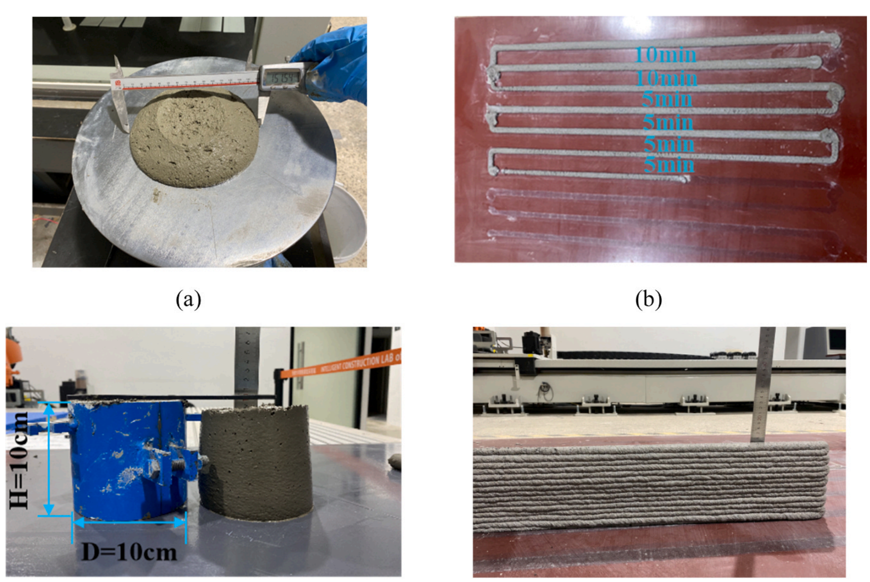
The additive manufacturing of the specimens was executed using a high-precision KUKA six-axis robotic arm (model KR210R00–2, Germany), integrated with a specialized extrusion system and a 3D printing control platform developed by Shanghai DAJIE® (as shown in Fig. 5). The control interface manages the extrusion rate by regulating the operating frequency of the rotary rod blades and the internal vibrator within the print head. A 1 m × 1 m logistics pallet served as the build plate for all printing operations. The digital workflow was orchestrated using customized Rhino software, with the Grasshopper plugin employed to generate the toolpath and define the critical printing parameters (Table 4). Based on the optimized mix design and verified research protocols, a 20 mm diameter circular nozzle was utilized. The printing speed was maintained at a constant 50 mm/s to ensure adequate layer adhesion and shape retention. These parameters were strategically selected to balance the extrudability of the 3DRRPC with the required buildability for multi-layer structural integrity.

2.3. Printability test

The printability of the developed 3DPRRC mixtures was systematically evaluated in accordance with the T/CBMF 184–2022 standard [49]. The testing program comprised two primary categories: the determination of the printable time window (via flowability and extrudability tests) and the assessment of structural buildability.

For the flow table test, freshly mixed paste was placed into a standard conical mold positioned on the flow table. After lifting the mold, the table was subjected to 25 drops. The spread diameters of the paste in two orthogonal directions were measured using a caliper, and the average value was taken as the flowability, as illustrated in Fig. 6(a). This parameter reflects the initial workability and deformation capacity of the mixture under dynamic conditions.

Extrudability was assessed by printing multiple extrusion filaments with a length of 2 m. The first three filaments were printed at intervals of 10 min, followed by subsequent filaments at 5-minute intervals. This process continued until continuous and stable extrusion could no longer



(a) (b) (c) (d)

Fig. 6. Printability test: (a) Flow table test, (b) Extrudability test: (c) Cylinder slump test, (d) Printed structure stability test.

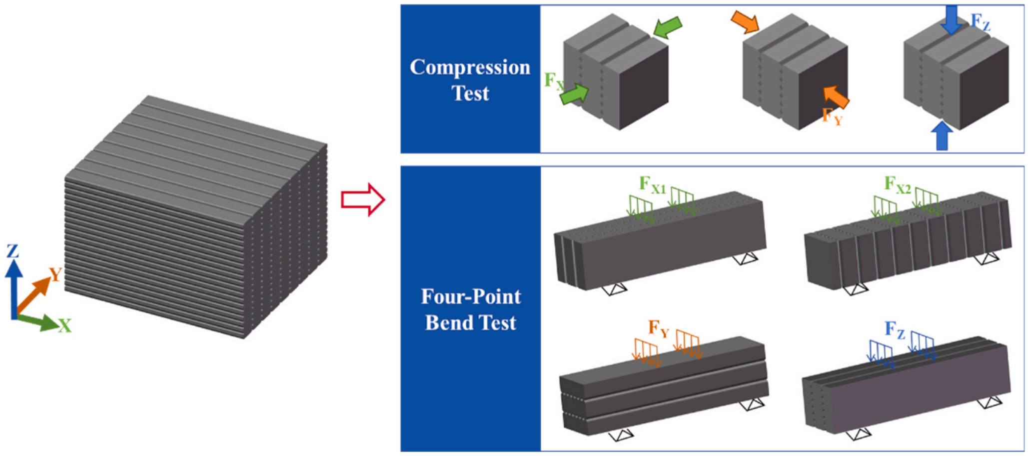


Fig. 7. Compressive and flexural strength test specimens and loading directions.

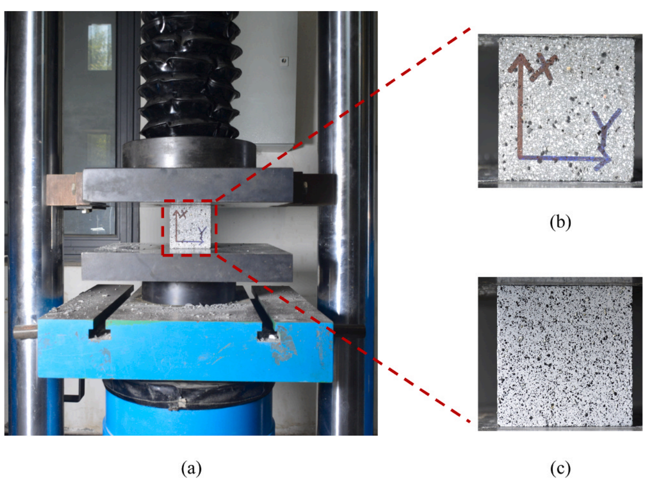


Fig. 8. Compressive testing equipment: (a) Compression testing machine, (b) Loading direction, (c) Speckle pattern for DIC.

be maintained, as shown in Fig. 6(b). The corresponding time at which extrusion failure occurred was defined as the printable time of the mixture, representing its effective working window for 3D printing.

The buildability defined as the ability of the material to resist self-weight and maintain structural integrity during layer-wise deposition was evaluated through a cylinder slump test and a multi-layer stability assessment.

The cylinder slump test utilized a rigid cylindrical mold (100 mm diameter, 100 mm height). The 3DPRRC was cast in three equal layers, leveled, and the mold was vertically extracted. The slump, or the vertical displacement from the original height to the specimen's apex, was measured after 30 s (Fig. 6c). This parameter serves as a critical indicator of its capacity for shape retention immediately after deposition.

Fig. 6(d) showed that structural stability was verified by printing a 15-layer vertical wall in a continuous, single-row stacking sequence. The dimensional accuracy of the final structure was quantified by measuring the vertical height deviation between the physical printed specimen and the digital design model. This test evaluates the cumulative deformation of the bottom layers under the increasing gravitational load of the subsequent layers, ensuring the mixture provides sufficient green strength for large-scale additive manufacturing.

2.4. Test sample configuration

The configuration of the test specimens was designed in accordance with T/CCPA 33–2022 [50]. To evaluate the directional mechanical performance, 3D-printed samples were prepared for uniaxial compressive strength tests along the X, Y, and Z directions, as well as for four-point bending tests along the X1, X2, Y, and Z orientations. Here, the X-direction corresponds to the printing direction, the Y-direction represents the inter-laminar (inter-strip) direction, and the Z-direction denotes the height direction. The printed bodies were then cut into test specimens: compressive specimens were cubic with edge lengths of 70.7 mm, and four-point bending specimens were rectangular prisms with dimensions of 70.7 mm × 70.7 mm × 300 mm (as shown in Fig. 7).

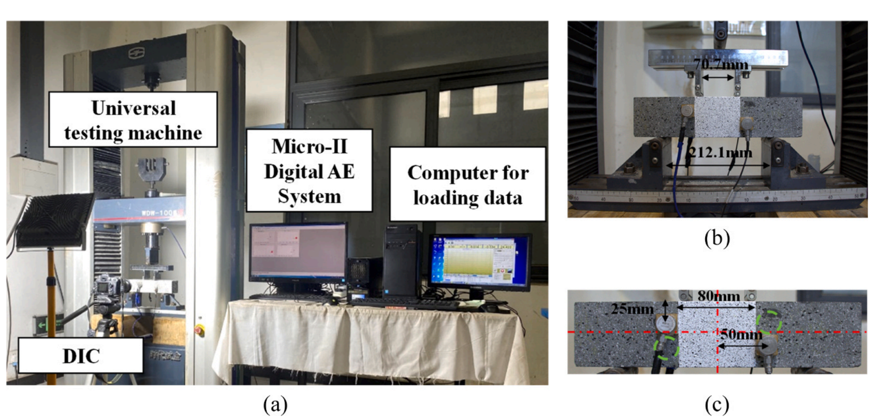


Fig. 9. Flexural testing, DIC, and AE testing systems: (a) Main equipment, (b) Loading dimensions, (c) Sensor placement.

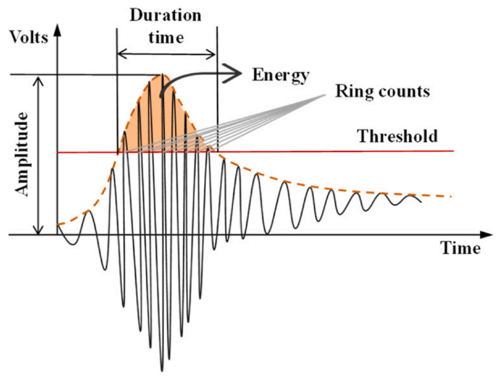


Fig. 10. Principle diagrams of AE.

2.5. Mechanical testing and evaluation methods

2.5.1. Mechanical property testing procedures

Compressive and flexural strength tests were carried out in accordance with T/CCPA 33–2022 [50] standards (as shown in Figs. 8 and 10). For compressive testing, each mix proportion of the 3D-printed specimens was evaluated in the X, Y, and Z directions, with six specimens tested per direction. For flexural testing, each mix proportion was tested in the X1, X2, Y, and Z directions, also with six specimens per direction. In contrast, three conventionally mold-cast specimens per mix proportion were selected for testing. All mechanical tests were conducted under displacement-controlled loading conditions, with loading rates of 2.5 kN/s for compressive strength and 0.083 kN/s for flexural strength. To ensure the accuracy and consistency of the experimental data, an outlier screening procedure was applied. Specifically, if the deviation of the maximum or minimum measured value exceeded 15% of the mean value, the corresponding data point was discarded. The remaining dataset was then re-evaluated using the same criterion. When at least three valid data points remained, their average value was taken as the representative strength for the corresponding group. Otherwise, the test results for that group were considered invalid. The compressive strengths of conventionally mold-cast specimens and 3D-printed specimens in the X, Y, and Z directions were denoted as $F_{C\text{-}CM}$, $F_{C\text{-}X}$, $F_{C\text{-}Y}$, and $F_{C\text{-}Z}$, respectively. The flexural strengths in the X1, X2, Y, and Z directions were denoted as $F_{F\text{-}CM}$, $F_{F\text{-}X1}$, $F_{F\text{-}X2}$, $F_{F\text{-}Y}$, and $F_{F\text{-}Z}$, respectively.

2.5.2. Digital image correlation (DIC) test

DIC is an important non-contact measurement technique for material deformation and provides intuitive data support for the analysis of material failure processes [51,52]. In compressive and flexural tests, DIC technology was employed to capture the full strain field of the specimens and conduct a comparative analysis of crack patterns between 3D-printed and conventionally cast concrete. The DIC system comprises a light source and a digital camera (Fig. 9a). For compressive specimens, the front surface was selected as the DIC monitoring area (Fig. 8c).

2.5.3. Acoustic emission (AE) test

When internal damage occurs in a material, it is accompanied by the rapid release of localized energy, thereby generating transient elastic waves. The AE technology can capture these elastic waves through sensors attached to the material surface. Fig. 10 shows a typical AE signal schematic, from which key parameters, such as energy, amplitude, and duration time, can be extracted as AE characteristics [53].

AE monitoring during the flexural testing of the specimens was performed using the SAMOSAE^{win} (Sensor-Based Acoustic Multi-Channel Operating System) multichannel monitoring system, a PC-based system developed by Physical Acoustics Corporation. Four R6α resonance-type sensors were mounted on the surface of each specimen (as shown in Fig. 8), with Vaseline applied to fill the gaps between the sensors and the specimen. In this test, the AE signals were amplified by 40 dB, and the AE threshold was set at 45 dB to ensure a high signal-to-noise ratio. The AE signals acquired by the sensors were transmitted to the Micro-II digital AE data acquisition system for recording, storage, and analysis. A displacement-controlled loading mode with a displacement rate of 0.1 mm/min was employed during testing.

Table 5
Printable time of fresh 3DPRRC with different RHA content.

Group	R0	R10	R20	R30
Printable time (min)	50	40	35	30

Table 6
Buildability of fresh 3DPRRC with different RHA content.

Group	R0	R10	R20	R30
Slump (mm)	15	13	10	11
Height loss (mm)	0	0	0	0

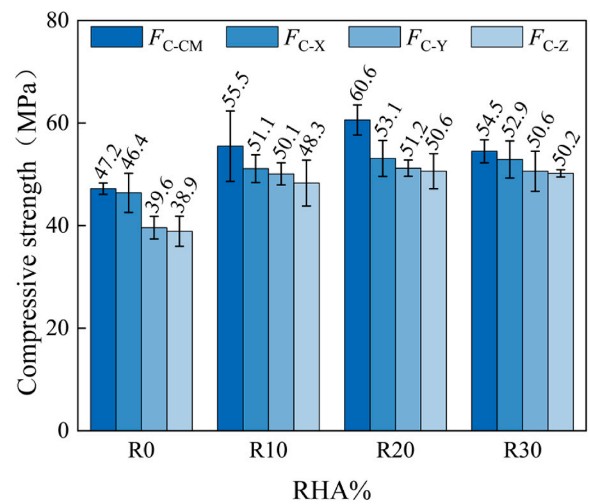


Fig. 11. Compressive strength.

3. Results and discussion

3.1. Evaluation of printability and buildability

Table 5 presents the printable times for mixtures incorporating varying proportions of RHA. As the RHA replacement ratio increased, the printability time exhibited a downward trend. This phenomenon is attributed to the large specific surface area and high surface activity of RHA, which accelerated the early hydration reaction of cement, thereby reducing the printable time. Simultaneously, the water absorption capacity of RHA reduces the operability of concrete. Although the printable time is shortened, R30 still maintains a printable time of 30 min, which remains sufficient for the printing process.

The buildability assessment (Table 6) revealed that the slump flow of concrete decreased with increasing RHA replacement ratio. However, the improved buildability of 3DPRRC is attributed to the water absorption capacity of RHA. Notably, none of the printed specimens exhibited height loss during the printing process. The height loss in printed specimens typically originates from the bottom layers, where insufficient stiffness and compressive stress from overlying layers cause the layer height to fall below the design value, while the width exceeds the design value. In contrast, the upper layers demonstrate an inverse trend: (1) the compressive stress on upper layers is smaller, and (2) the upper layers have a longer opening time compared to the lower layers.

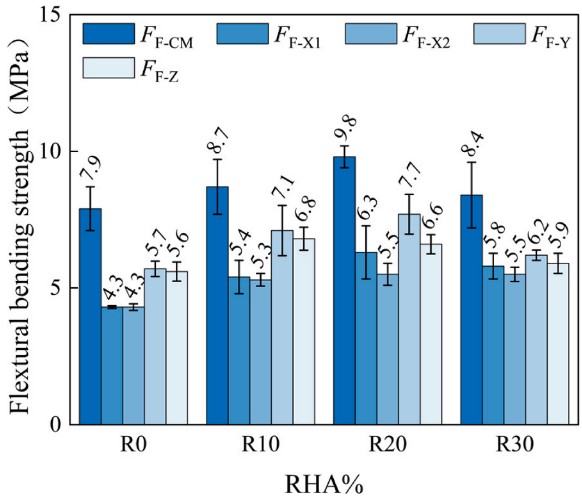


Fig. 12. Flexural strength.

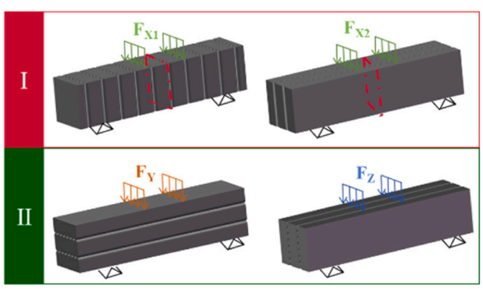


Fig. 13. Classification of flexural test specimens.

As cement hydration progresses, the upper layers develop sufficient strength to maintain their shape, compensating for height loss by sacrificing width.

3.2. Evaluation of mechanical properties and anisotropic behavior in 3DPRRC

The experimental results for the compressive and flexural strengths are illustrated in Figs. 11 and 12, respectively. A comparative analysis across varying RHA replacement ratios reveals that the mechanical properties of 3DPRRC are significantly enhanced upon the substitution of cement with RHA. Within the same mixed design, the compressive and flexural strengths of the mold-cast specimens are higher than those of the 3D-printed specimens due to the presence of weak interlayer interfaces in the printed samples. The compressive strength hierarchy is as follows: $F_{C\text{-}CM} > F_{C\text{-}X} > F_{C\text{-}Y} > F_{C\text{-}Z}$, whereas the flexural strength hierarchy is as follows: $F_{F\text{-}CM} > F_{F\text{-}Y} > F_{F\text{-}Z} > F_{F\text{-}X1} > F_{F\text{-}X2}$.

Among the specimens, the R20 mix exhibited the highest compressive and flexural strengths. Compared with the R0 mix, the compressive strength in $F_{C\text{-}CM}$, $F_{C\text{-}X}$, $F_{C\text{-}Y}$ and $F_{C\text{-}Z}$, increased by 24.9%, 16.0%, 31.5%, and 32.8%, respectively. Similarly, the flexural strengths in $F_{F\text{-}CM}$, $F_{F\text{-}X1}$, $F_{F\text{-}X2}$, $F_{F\text{-}Y}$ and $F_{F\text{-}Z}$ improved by 24.4%, 46.6%, 28.1%, 35.6%, and 18.6%, respectively. These results demonstrate the effectiveness of RHA in enhancing the strength of the weakly loaded directions in both compressive and flexural performances.

Notably, even at a high cement replacement level of 30% by mass, the mechanical properties of the R30 mix remained superior to those of the R0 baseline. These findings indicate that ground RHA serves as a highly viable and sustainable cementitious substitute, facilitating a reduction in clinker consumption and a marked enhancement in the structural performance of 3D-printed rubberized systems.

Based on the relationship between the extrusion direction of the concrete strands and the orientation of the applied load (parallel or perpendicular), this study classified 3D-printed flexural test specimens into two categories (Fig. 13). In the first category of specimens (X1 and X2 directions), the printing direction of the concrete extrudates was consistent with the loading direction. Consequently, cracks may propagate continuously along the weak interfacial layers across the entire specimen, and the structural strength is primarily determined by the interfacial bond strength. In the second category of specimens (Y and Z directions), the printing direction of the extruded concrete strands was orthogonal to the loading direction. Consequently, crack propagation requires the crossing of multiple strands, and the structural strength is primarily governed by the matrix strength. Therefore, the flexural strengths of similar specimens were relatively consistent, and the flexural strength of the second type of specimen was higher than that of the first type.

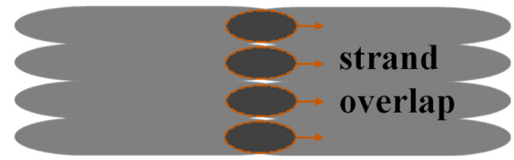


Fig. 14. Overlapping phenomenon between 3D-printed concrete strands.

Table 7
Parameter S for calculating the anisotropy coefficients of different mechanical strengths.

I_a	S	n
For the compressive anisotropy coefficient	$\{i\|X, Y, Z\}$	3
For the flexural anisotropy coefficient	$\{i\|X1, X2, Y, Z\}$	4
For the first category flexural anisotropy coefficient	$\{i\|X1, X2\}$	2
For the second category flexural anisotropy coefficient	$\{i\|Y, Z\}$	2

Table 8
Mechanical anisotropy coefficient.

	Group	R0	R10	R20	R30
I_a	Compressive	3.38	1.16	1.07	1.19
	Flexural	0.68	0.81	0.79	0.25
	Category I	0	0.05	0.4	0.15
	Category II	0.05	0.15	0.55	0.15

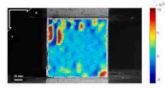
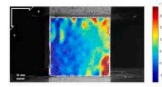
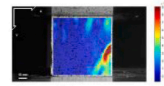
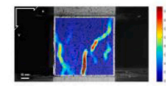
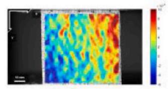
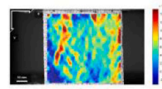
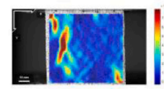
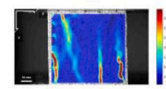
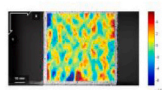
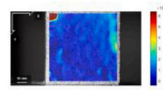
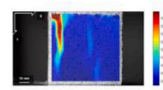
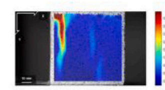
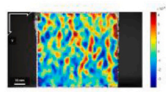
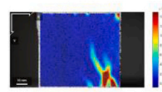
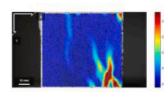
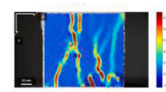
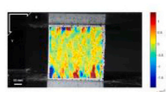
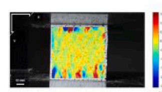
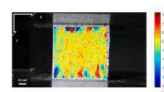
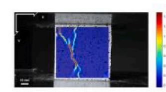
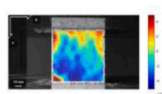
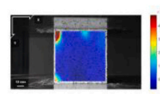
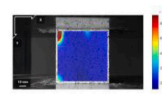
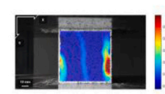
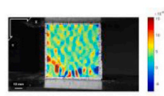
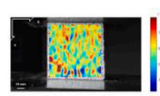
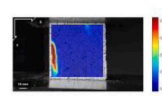
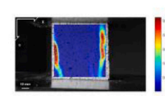
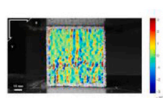
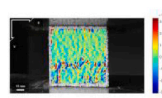
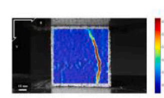
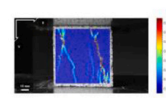
During 3D printing, overlapping regions between concrete strands occur due to compression from upper layers (Fig. 14), which explains why the flexural strength in the X1 direction exceeds that in the X2 direction, and the Y-direction flexural strength is greater than the Z-direction flexural strength.

Due to the layer-by-layer stacking process in 3D printing, 3DPC exhibits significant mechanical anisotropy. To quantitatively assess the impact of the RHA replacement ratio on the anisotropic behavior of 3DPRC, the anisotropy coefficient$I_a$proposed by Wang Li et al. [54] is referenced as a characterization index, which are given by:

$$I_a = \sqrt{\frac{1}{n}\sum_{i\in S}\left(f_i - f_{avg}\right)^2} \tag{1}$$

Wheref_{avg}represents the average mechanical strength of the specimen across different loading directions.$f_i$denote the mechanical strength of the 3DPC specimen in different loading directions. Table 7 lists the selection scheme for parameter S used in calculating the anisotropy coefficients of different mechanical strengths. n denotes the number of loading directions.$I_a$mechanical denotes the anisotropy coefficient, which is defined as the standard deviation between the strength in a single direction and the mean strength along the loading direction. For isotropic materials, the anisotropy coefficient$I_a$is equal to 0, meaning

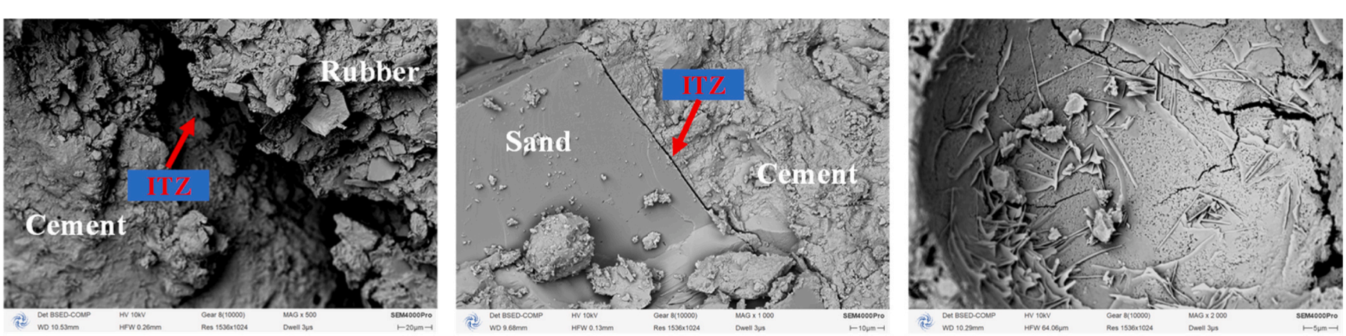
Table 9
DIC strain maps of crack propagation in R0 and R20 specimens during compression testing.

Group		25% Load	50% Load	75% Load	100% Load
R0	Cast				
	X				
	Y				
	Z				
R20	Cast				
	X				
	Y				
	Z				

that the smaller the value, the less significant the anisotropic difference.

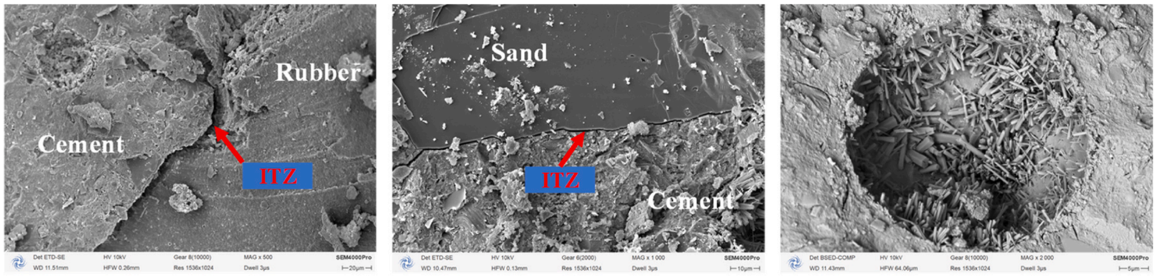
Table 8 presents the calculation results, which characterize the improvement in the mechanical anisotropy of 3DPRC caused by the incorporation of RHA. Under compressive loading, the mechanical anisotropy of 3DPRC is significantly reduced. Furthermore, the substitution of 20% cement with RHA results in a significantly reduced anisotropy coefficient in 3DPRC.

In the flexural loading direction, the mechanical anisotropy of 3DPRC slightly increased. Among them, the R30 specimen exhibits the lowest anisotropy coefficient. This improvement is attributed to the promotion of cement hydration at the edges of the extruded layers by RHA, which enhances the interfacial bond strength. However, the high replacement level of RHA leads to the absorption of a large amount of free water, resulting in incomplete hydration of cement within the strand matrix and, consequently, no significant enhancement in the matrix strength. As a result, the interfacial bond strength of the R30 specimen is comparable to the matrix strength of the extruded strands, leading to reduced mechanical anisotropy. Given that the flexural failure modes of specimens of the same type are consistent, their anisotropy coefficients are relatively low.



(a) (b) (c)

Fig. 15. The SEM microstructure of the R0 specimen: (a) ITZ between cementitious matrix and rubber, (b) ITZ between cementitious matrix and sand, (c) Cement hydration products.



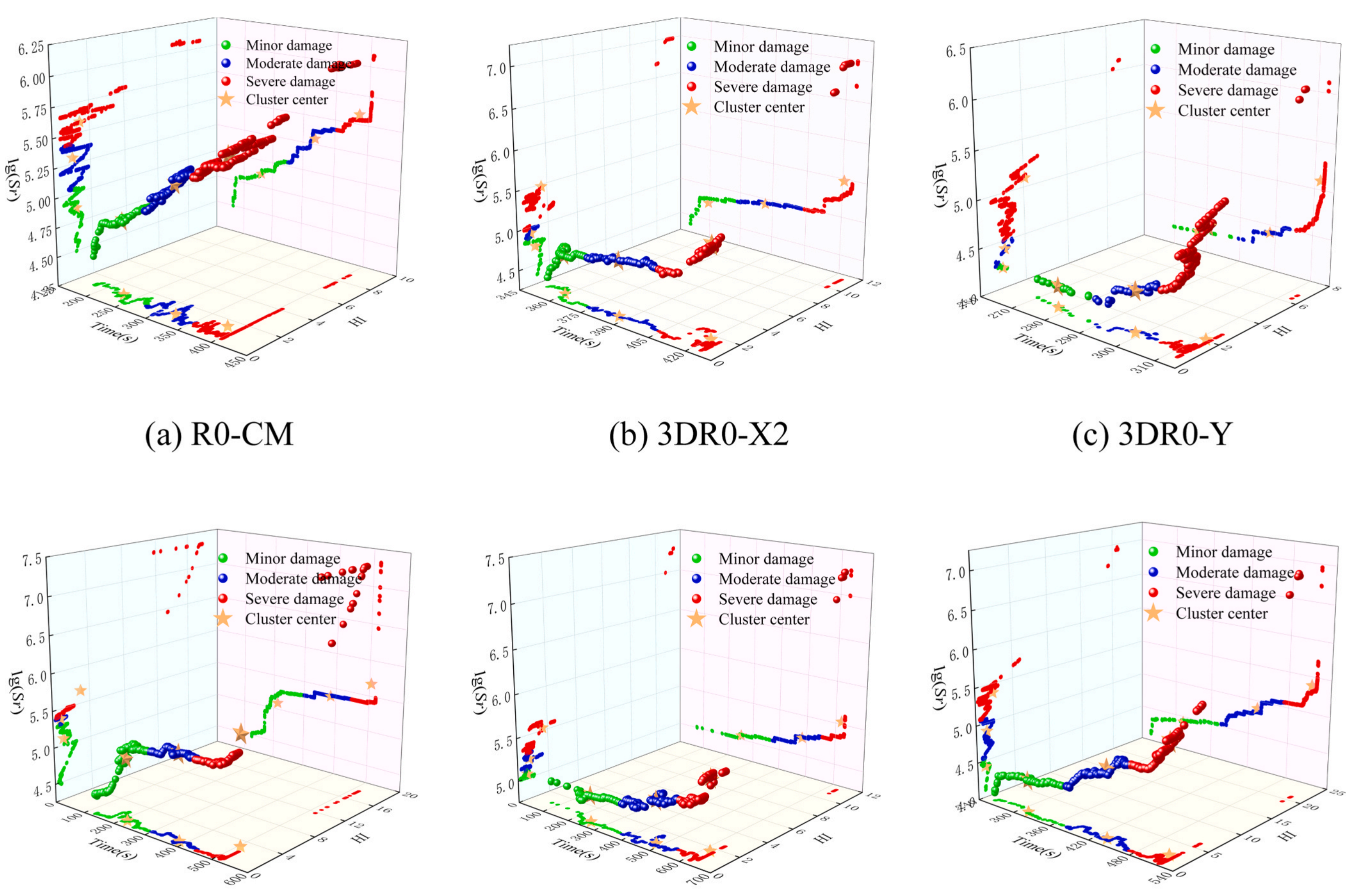
(a) (b) (c)

Fig. 16. The SEM microstructure of the R20 specimen: (a) ITZ between cementitious matrix and rubber, (b) ITZ between cementitious matrix and sand, (c) Cement hydration products.

The incorporation of RHA, combined with a classification approach for flexural specimens, can mitigate the anisotropy of 3D-printed concrete to a certain extent. This renders the material properties more consistent with the isotropic assumption, thereby holding potential positive implications for improving design accuracy and enhancing construction controllability.

3.3. Digital image correlation (DIC) analysis

As illustrated in Table 9, the DIC analysis of R0 and R20 specimens during compression testing reveals distinct crack propagation patterns. While the Z-direction specimen exhibited diagonal cracking, the X- and Y-direction specimens failed via interfacial debonding, with cracks aligned parallel to the loading direction. The R20 mix demonstrated reduced crack density in the X- and Y-directions owing to RHA-induced interfacial strengthening. These findings provide insights into the anisotropic mechanical behavior of 3DPRRC under compression. The X- and Y-direction specimens were segmented into multiple short columns by vertical cracks, allowing continued load-bearing capacity after initial cracking. However, the short columns in the Y-direction specimens contained inter-strand weak interfaces, leading to a reduced load transfer efficiency compared to that in the X-direction specimens. This explains the anisotropic compressive behavior of 3DPC, where weak interfacial zones dominate the failure mechanisms in specimens loaded in the X- and Y-directions, whereas cracks in the Z-direction specimens must traverse through the concrete bulk, exhibiting a fracture pattern similar to that of mold-cast specimens. These DIC findings provide robust empirical evidence for the role of RHA in bridging the



(a) R0-CM (b) 3DR0-X2 (c) 3DR0-Y

(d) R20-CM (e) 3DR20-X2 (f) 3DR20-Y

Fig. 17. *HI*-lg(*Sr*) Time-History scatter plot.

Table 10
Cluster center of signal intensity.

Group	Minor damage stage		Moderate damage stages		Severe damage stages	
	HI	lg(*Sr*)	*HI*	lg(*Sr*)	*HI*	lg(*Sr*)
R0-CM	1.13	4.87	1.03	5.30	1.47	5.58
3DR0-X2	1.10	4.74	0.84	4.93	1.62	5.35
3DR0-Y	0.10	4.22	1.10	4.41	2.05	5.09
R20-CM	1.04	5.09	0.81	5.37	3.04	5.67
3DR20-X2	0.59	5.05	0.74	5.22	1.67	5.53
3DR20-Y	0.10	4.42	1.35	4.89	2.29	5.38

performance gap caused by manufacturing-induced anisotropy.

3.4. SEM analysis

Figs. 15 and 16 illustrates the microstructural characteristics of the R0 and R20 specimens. As observed in Figs. 15(a-b) and 16(a-b), the rubber-cement interfacial transition zone (ITZ) is significantly weaker than the sand-cement ITZ. This discrepancy is attributed to the poor interfacial bonding between the hydrophobic rubber particles and the cement matrix, which explains the reduction in mechanical properties of the 3D-printed rubberized concrete (3DRPC) upon rubber incorporation. Furthermore, Figs. 15(c) and 16(c) shows the cement hydration products in the micropores of R0 and R20 specimens, indicating that the addition of RHA promotes the cement hydration, generates more hydration products, fills the micropores, and makes the internal microstructure of R20 specimens denser, thus improving the mechanical properties of R20 specimens.

3.5. AE characteristic parameter analysis

Due to the limitation of article length, this study selects the X2 and Y directions of the mold-cast and 3D-printed specimens with two mix proportions, R0 and R20, as representatives for further analysis.

3.5.1. Signal intensity analysis

The damage state of concrete structures can be characterized by AE signal intensity analysis. This type of analysis includes images of the AE historical index (*HI*) and logarithmic severity (lg(*Sr*)), which are used to track the temporal and cumulative behavior of AE events during loading. In this study, the AE amplitude and duration are considered as intensity parameters to demonstrate that AE intensity analysis is capable of predicting the trend of AE data evolution. The method for AE signal intensity analysis is based on the approach described in literature [55,56].

Fig. 17 shows the *HI*-lg(*Sr*) time-history scatter plots for R0 and R20 specimens. The mutation points in the *HI* time-history scatter plot (X-Y plane) are consistent with those in the lg(*Sr*) time-history scatter plot (X-Z plane). Therefore, the *HI*-lg(*Sr*) (Y-Z plane) can be used in a coordinated manner to characterize the damage state of the specimens. In literature [40], *HI*-lg(*Sr*) scatter plots are clustered to classify damage states: 1) Low *HI* and low lg(*Sr*) values indicate minor damage. 2) Low *HI* and high lg(*Sr*) values correspond to moderate damage. 3) High *HI* and high lg(*Sr*) values indicate severe damage. However, these studies do not account for the time-dependent accumulation of damage. The damage state of the specimen is closely related to the loading time, as the AE event density increases over time within the specimen under loading.

The K-means clustering algorithm initializes k centroids and alternately performs sample assignment (partitioning each point into the cluster of its nearest centroid) and centroid update (recalculating the mean of points within each cluster as new centroids), thereby minimizing the within-cluster sum of squared errors to iteratively optimize the clustering results. In this study, K-means clustering is applied to the *HI*-lg(*Sr*) time-history scatter plots, dividing the data into three clusters that represent the minor, moderate, and severe damage stages during the loading process. Table 10 presents the cluster centers in terms of *HI* and lg(*Sr*) values.

In the minor damage stage, the *HI*-lg(*Sr*) time-history scatter plots of R0-CM and R20-CM specimens exhibit similar trends, which can be attributed to the poor bonding between rubber particles and the cement matrix. In both R0-CM and R20-CM specimens, cracks initiate at the weak interface between rubber and cement. However, in the moderate and severe damage stages, the scatter plot of R20-CM is smoother, indicating a slower crack propagation rate compared to R0-CM. Additionally, the *HI* and lg(*Sr*) time-history scatter plots of R20-CM show more peaks than R0-CM, suggesting that fewer large crack events occur in R20-CM during loading. For 3D-printed specimens, the *HI*-lg(*Sr*) time-history scatter plots follow the same trend under the same loading direction. In the Y-direction loading, cracks must traverse the entire concrete strand, which is a more energy-intensive process. As a result, the *HI*-lg(*Sr*) time-history scatter plot in the severe damage stage of Y-direction loading shows a similar trend to that of cast concrete, as the damage mechanism becomes comparable in terms of crack propagation and energy release.

In the minor damage stage, the lg(*Sr*) value of the cluster center for R20 specimens is lower than that of R0 specimens, while the *HI* value is not higher than that of R20 mix specimens, indicating that crack initiation in R20 is accompanied by stronger AE events. The incorporation of RHA densified the micro-pores within the specimens, filling the capillary voids and refining the pore structure. In the moderate damage stage, except for the 3D-printed Y-direction specimens, the trends of *HI* and lg (*Sr*) values for other specimens are consistent with those in the minor damage stage. For the Y-direction specimens, the cracks propagated from the interlayer to the matrix strips, resulting in a significant increase in signal intensity variation. Due to the strengthening effect of RHA, Both the cluster center *HI* and lg(*Sr*)value for 3DR20-Y exceed those of 3DR0-Y. In the severe damage stage, the cluster centers of R20 specimens show both higher *HI* and lg(*Sr*) values compared to R0 specimens, suggesting that RHA provides additional structural safety reserve by reducing the likelihood of catastrophic failure.

3.5.2. Ib-value analysis

The b-value analysis was first introduced in seismology for the analysis of seismic waves. In AE testing, the received AE waves are similar to seismic waves, and the internal material failure in mortar specimens can be considered as a micro-seismic event. Therefore, the b-value analysis can also be applied to AE signal analysis [57]. To better characterize the statistical properties of AE event amplitudes, and in consideration of the limitations of the two commonly used b-value estimation methods-least squares and maximum likelihood methods, Shiotani et al. [58,59] proposed an improved b-value analysis method, known as the Ib-value (Improved b-value), which is particularly suitable for the analysis of AE signals in brittle materials such as rocks and cementitious composites. The calculation formula is shown in Eq. (2):

$$Ib = \frac{\lg N(\mu - \alpha_1\sigma) - \lg N(\mu + \alpha_2\sigma)}{(\alpha_1 + \alpha_2)\sigma} \tag{2}$$

Where, $\sigma$and$\mu$are the standard deviation and the mean of the amplitude statistics within the AE event group, respectively.$\alpha_1$and$\alpha_2$are correlation coefficients related to the amplitude of AE events.$N(\mu - \alpha_1\sigma)$is the number of AE events with amplitude greater than$\mu - \alpha_1\sigma$. $N(\mu + \alpha_2\sigma)$is the number of AE events with amplitude greater than$\mu + \alpha_2\sigma$. Studies have shown that the amplitude and hit count of AE events follow a normal distribution, and the calculation process of the Ib-value reflects these characteristics of normal distribution. Ib-value varies even under very minor damage events, due to its high sensitivity. Typically, the statistical parameter N is set to 50 hits ($N = 50$) for analysis.

Generally, micro-cracks release a larger number of AE signals with smaller amplitudes, whereas macro-cracks release fewer signals with

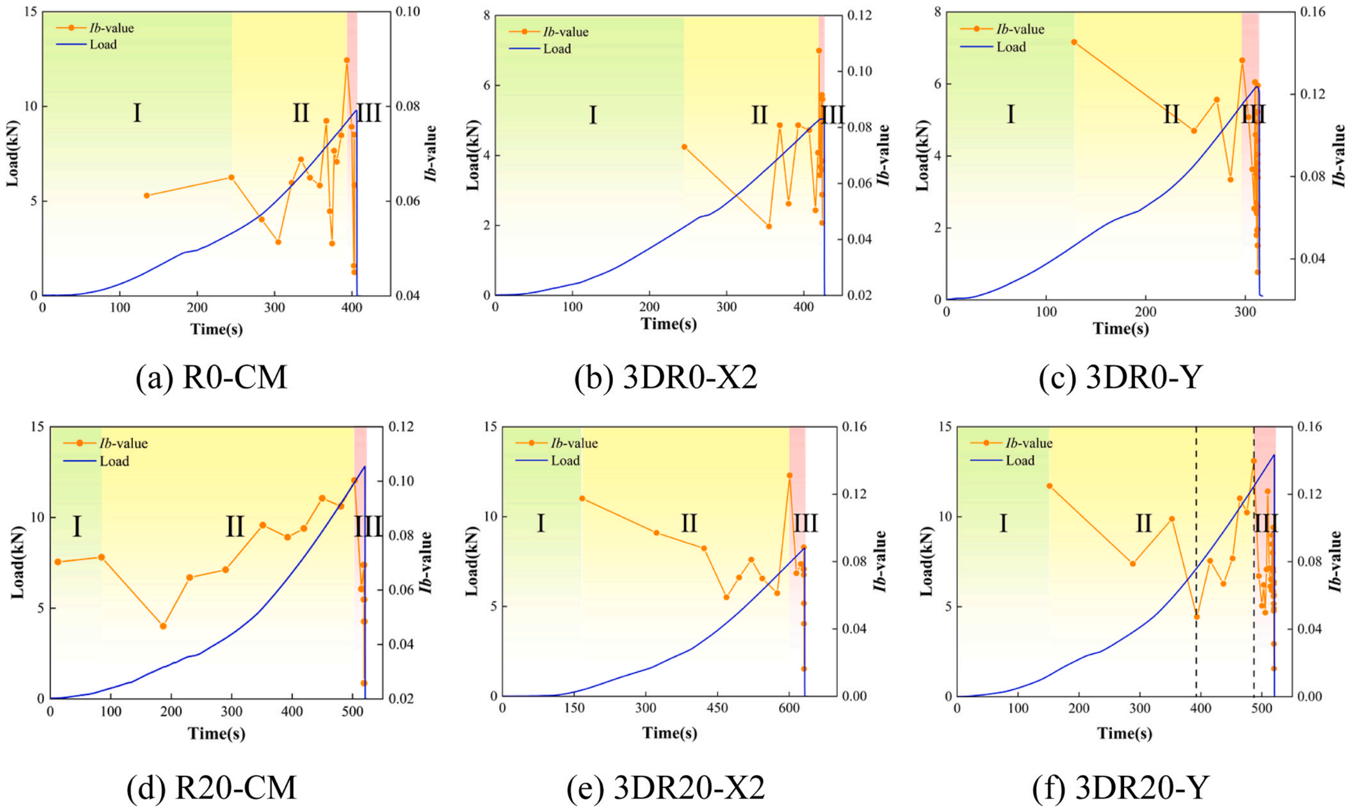


Fig. 18. The Ib-value distribution.

larger amplitudes. The Ib-value, calculated from Eq. (2), can be used to characterize the size of cracks within the mortar: a higher Ib-value indicates smaller crack lengths and widths within the mortar, corresponding to the development of micro-cracks; conversely, a lower Ib-value indicates larger crack lengths and widths, corresponding to the formation of macro-cracks. The variation of the Ib-value and the amplitude distribution for the three mortar specimens during the four-point bending test are illustrated in Fig. 18 In the initial loading stage, the Ib-value of the mold-cast specimens R0 and R20 first increase and then decrease, corresponding to the compaction phase of the concrete. In contrast, the Ib-value of the 3D-printed specimens do not show an increasing phase, but rather decrease directly, indicating that the layer-to-layer and strand-to-strand weak interfaces in 3D-printed specimens act as initial damage sites, and cracks can propagate directly along these interfaces.

In the middle loading stage, the Ib-value of the R0 specimen fluctuates significantly, and the fluctuations become more pronounced as the load increases, suggesting that cracks initiated in R0 propagate rapidly within the specimen. Meanwhile, the Ib-value of the R20 specimen shows an increasing trend with much smaller fluctuations than R0, indicating that the addition of RHA improves the specimen strength and helps resist crack propagation.

Similarly, the Ib-value of the 3D-printed specimens R20-X2 and R20-Y also exhibit smaller fluctuations compared to R0-X2 and R0-Y, which suggests that the inclusion of RHA enhances the interlayer and inter-strand bonding strength in 3D-printed concrete. It is worth noting that the Ib-value of the 3DR20-Y specimen shows an increasing trend, a behavior similar to that observed in the cast specimen, as indicated by the dashed line interval in Fig. 18(f). This is due to the fact that crack propagation in 3DR20-Y must traverse through the concrete strands, leading to a damage evolution pattern similar to that of cast concrete.

Prior to the ultimate failure of the specimens, the Ib-value reached its peak and then underwent a sharp decline accompanied by violent oscillations. This phenomenon indicates that, within a very short time

Table 11
Minimum and maximum Ib-value analysis of R0 and R20 specimens.

Group	Ib_{min}	Ib_{max}	Ib_{max}-Ib_{min}
R0-CM	0.045	0.090	0.045
3DR0-X2	0.045	0.107	0.062
3DR0-Y	0.034	0.137	0.077
R20-CM	0.046	0.100	0.075
3DR20-X2	0.051	0.128	0.077
3DR20-Y	0.038	0.137	0.099

interval, a large number of micro-cracks and macro-cracks initiated and coalesced into a dominant macro-crack, leading to the final failure of the specimen. Regarding oscillation intensity, the R20 specimen displayed milder fluctuations than the R0 specimen, while the 3D-printed specimens showed more severe oscillations than the cast counterparts. These findings suggest a lower fracture severity for R20 compared to R0, and a higher fracture severity for 3D-printed specimens relative to cast ones. Table 11 lists the maximum and minimum Ib-value (Ib_{max} and Ib_{min}) and their ranges during the failure phase. The results indicate that superior interfacial bonding constrains crack development, leading to slower propagation rates and consequently higher Ib-value. For the cast specimens, the Ib_{max} followed the trend R20 > R0. In contrast, for the 3D-printed specimens, the trend was R20 > R0. This indicates that the incorporation of RHA contributes to enhancing the strength of the concrete strip matrix and its interfacial bonding strength.

Although the variation amplitude of the Ib-value during fracture was significantly larger for the R20 specimen compared to the R0 specimen, the oscillation intensity was lower in the R20 specimen. This suggests that while the R20 specimen released higher energy upon failure, it exhibited fewer internal cracks. Consequently, the energy required to generate a unit crack in the R20 specimen is higher than that in the R0 specimen. Similarly, the energy required to generate a unit crack in the 3D-printed specimens is lower than that in the cast specimens. The Ib-

Table 12
Equipment and carbon emission factors for RHA and C in the "Production-Transportation" stage [60].

Cementitious materials	Stage	Equipment	Carbon emission factor
RHA	Transportation (680 km)	Heavy-duty Diesel Trucks	0.129 kg CO_2e/(t·km)
	Grinding (50 min)	Small Ball Mill*	221.075 kg CO_2e/t
C	Production	-	735 kg CO_2e/t
	Transportation (40 km)	Medium-duty Trucks	0.115 kg CO_2e/(t·km)

* The effective loading capacity (G) of the small ball mill is 5 kg. The rated power of the equipment (P) is 1.5 kW. The grinding time (T) is 5/6 h. The carbon emission factor of the North China power grid (e_0) is 0.8843 kg CO_2/kWh. The carbon emission factor (e_1) for grinding RHA is calculated as: $e_1 = \frac{P \times T \times e_0 \times 1000}{G}$.

value effectively explain why the strength of the R20 specimen is higher than that of the R0 specimen, and why the strength of the 3D-printed specimens is lower than that of the cast specimens.

3.6. Environmental impacts of the 3DCRRP

To quantify the carbon emission reduction achieved by replacing cement with RHA, we conducted a comparative analysis of the carbon emissions associated with the materials used in 1 m3 of R0 and R20 mixtures during their production and transportation phases. As the R0 and R20 mixtures differed only in the binder content (the water reducer accounted for less than 0.3% of the total material weight; therefore, the difference in water reducer dosage was negligible). The carbon emission inventory was compiled in strict accordance with the Chinese Standard [60]. In accordance with Section 6.2.5 of Standard [60], as rice husk is considered a high-quality biomass, the carbon emissions generated from its combustion into RHA were excluded from the calculation. Consequently, for RHA, only the carbon emissions from the transportation and grinding stages were calculated. Table 12 presents the equipment and carbon emission factors for the production and transportation stages of both RHA and cement (C).

Fig. 19 presents the carbon emission calculation results for the cementitious materials required for R20 and R0 during the "Production-Transportation" phase. Fig. 19(a) illustrates the carbon emission comparison between 1 metric ton of RHA and cement. Under equivalent mass conditions, the carbon emissions of RHA are only 41.75% of those of cement, demonstrating a significant carbon reduction advantage. However, the high carbon contribution rates of RHA in these two stages are attributed to inter-provincial transportation and secondary grinding conducted in the laboratory to enhance reactivity after calcination (a high-energy consumption process).

Given the widespread distribution of rice husks in the Tianjin region and the energy-saving potential of industrial-grade ball mills, the carbon emissions associated with RHA production and transportation are expected to decrease further if "local sourcing" and scale grinding are achieved in the future. Fig. 19(b) shows a comparison of the carbon emissions of cementitious materials for the R0 and R20 mix designs per 1 cubic meter during the "Production-Transportation" stage. The R20 mix design exhibited lower carbon emissions, with a reduction of 11.65% compared to R0. Therefore, substituting cement with RHA enhances the strength of 3D-printed rubber concrete and achieves low-carbon performance, effectively promoting its application in high-load-bearing and sustainable development scenarios.

4. Conclusions

To address the issues of strength degradation in 3DPRC and excessive cement usage in 3D-printed concrete, this study replaced a portion of the cement in 3DPRC with RHA to enhance its mechanical properties. The printability, compressive strength, and flexural strength of 3DPRRC were investigated. Additionally, the damage evolution of 3DPRRC was monitored using AE techniques. The following conclusions were drawn:

(1) The internal porous structure and high pozzolanic activity of RHA contribute to the improvement of the buildability of 3DPRC, and its printability is sufficient to meet the printing requirements. When the RHA replacement rate is 20%, it significantly enhances the mechanical properties of 3DPRC. The addition of RHA shows a notable improvement in the mechanical performance of the weaker loading directions in 3DPRC, indicating that RHA can effectively reduce the mechanical anisotropy of 3DPRC.
(2) The crack patterns at failure differed between the 3D printed and mold-cast specimens. The mold-cast specimens failed primarily through diagonal cracking, whereas in the 3D printed specimens, cracks in the X- and Y- directions propagated along the weak interfacial layers, resulting in vertical crack patterns. In the Z-direction specimens, the crack must traverse through the printed

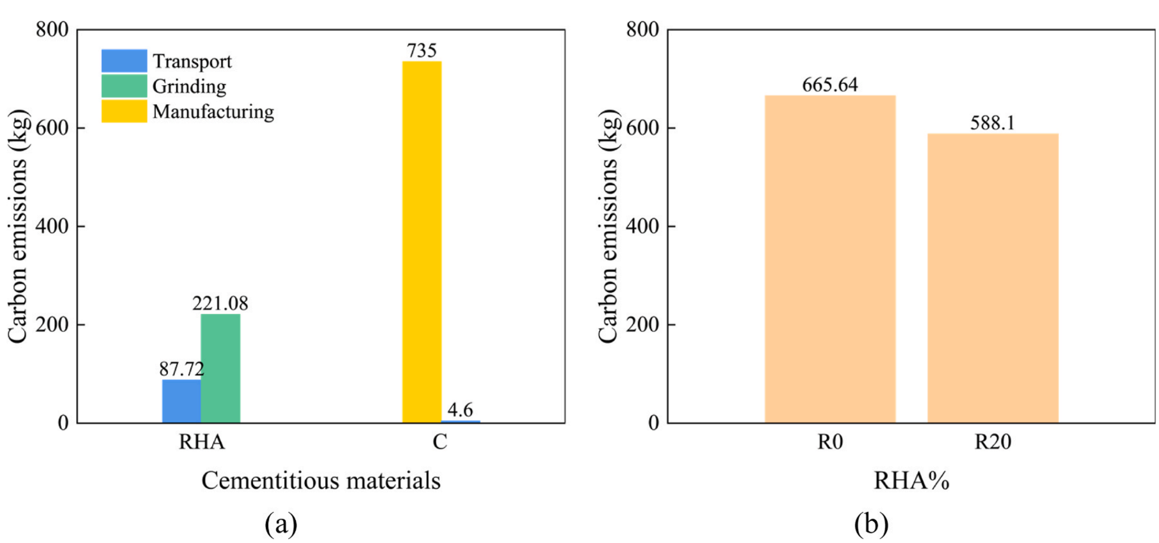


Fig. 19. Carbon emission calculation results for cementitious materials during the "Production-Transportation" stage: (a) Emissions for 1 t of RHA and C; (b) Emissions for cementitious materials of R0 and R20 mix designs per 1 m^3.

concrete strands, leading to a crack pattern similar to that of the mold-cast specimens.

(3) AE intensity and Ib-value analyses show that mold-cast specimens and 3D-printed specimens differ significantly, whereas 3D-printed specimens with the same loading orientation behave similarly. The slower crack growth in R20 specimens confirms that RHA-enhanced cement improves the weak interlayer bonding in 3DRPC.

(4) From a sustainability perspective, the carbon emissions of RHA during the "Production-Transportation" stage are only 41.75% of those of cement, indicating that RHA still holds significant potential for carbon reduction. A comparison of carbon emissions between the R0 and R20 binder mixtures at the same mass in the "Production-Transportation" stage reveals that the R20 mixture decreased by 11.65% compared to the R0 mixture.

This research demonstrates the technical feasibility of coupling agricultural waste (RHA) with industrial byproduct (rubber) within a digital fabrication framework. The findings provide a robust reference for the optimization of sustainable, data-driven construction materials that harmonize mechanical performance with environmental stewardship. Finally, given the excellent thermal and acoustic physical properties of 3DRRPC, it is highly suitable for 3D printing applications in highly insulated housing and road sound barrier projects.

CRediT authorship contribution statement

Pengpeng Jia: Writing – original draft, Methodology, Investigation, Data curation. **Qinghua Han:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Jie Xu:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of Competing Interest

No

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Data availability

Data will be made available on request.

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